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HYBRID ARC WELDING SYSTEM

Abstract

A hybrid arc welding system includes a waveform generator comprising a switching type power converter. A welding torch is operatively connected to the switching type power converter to receive electrical energy and produce an electric arc for welding. The system includes an engine-generator and a rectifier connected to receive an output of the engine-generator and supply electrical energy to the switching type power converter. A battery is connected to receive an output of the rectifier and supply electrical energy to the switching type power converter. A bidirectional battery charge-discharge circuit is connected between the rectifier and the switching type power converter and comprises a first and second electronic switches. A controller is operatively connected to the bidirectional battery charge-discharge circuit to control operations of the first and second electronic switches such that the bidirectional battery charge-discharge circuit is configured to selectively conduct charging current from the rectifier to the battery and selectively conduct welding current from the battery to the switching type power converter.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to a hybrid arc welding system having both an engine driven generator and a battery for supplying welding current to a welding torch.

Description of Related Art

[0002] Engine driven welders include an internal combustion engine, such as a diesel, gasoline or liquefied petroleum gas (LPG) engine. The engine drives a generator, and the generator supplies electrical power to a welding power supply that outputs a welding current and voltage. A hybrid engine welder includes a storage battery that can supply additional electrical power to the welding power supply to generate the welding current and voltage. The storage battery is charged by the generator when other current demand from the generator is low. The charging circuitry for the battery adds costs and space requirements to the engine driven welder and it would be desirable to minimize the number of components and the cost of such charging circuitry. A hybrid welding system is discussed in U.S. Pat. No. 10,166,624, titled “Hybrid Welding Supply”, and issued on Jan. 1, 2019, the entirety of which is incorporated by reference herein.

BRIEF SUMMARY OF THE INVENTION

[0003] The following summary presents a simplified summary in order to provide a basic understanding of some aspects of the devices, systems and/or methods discussed herein. This summary is not an extensive overview of the devices, systems and/or methods discussed herein. It is not intended to identify critical elements or to delineate the scope of such devices, systems and/or methods. Its sole purpose is to present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0004] In accordance with one aspect of the present invention, provided is a hybrid arc welding system. The hybrid arc welding system includes a welding waveform generator comprising a switching type power converter. A welding torch is operatively connected to the switching type power converter to receive electrical energy from the switching type power converter and produce an electric arc for welding. The system further includes an engine-generator and a rectifier connected to receive an output of the engine-generator and supply electrical energy to the switching type power converter. A battery is connected to receive an output of the rectifier and supply electrical energy to the switching type power converter. A bidirectional battery charge-discharge circuit is connected between the rectifier and the switching type power converter. The bidirectional battery charge-discharge circuit comprises a first electronic switch and a second electronic switch. A controller is operatively connected to the bidirectional battery charge-discharge circuit to control operations of the first electronic switch and the second electronic switch such that the bidirectional battery charge-discharge circuit is configured to selectively conduct charging current from the rectifier to the battery and selectively conduct welding current from the battery to the switching type power converter.

[0005] In accordance with another aspect of the present invention, provided is a hybrid arc welding system. The hybrid arc welding system includes a welding waveform generator comprising a switching type power converter. A welding torch is operatively connected to the switching type power converter to receive electrical energy from the switching type power converter and produce an electric arc for welding. The system further includes an engine-generator and a rectifier connected to receive an output of the engine-generator and supply electrical energy to the switching type power converter. A battery is connected to receive an output of the rectifier and supply electrical energy to the switching type power converter. A bidirectional buck-boost circuit is connected between the rectifier and the switching type power converter. The bidirectional buck-

boost circuit comprises a battery charging switch without a separate diode connected in parallel with the battery charging switch, and a battery discharging switch without a separate diode connected in parallel with the battery discharging switch. A controller is operatively connected to the bidirectional buck-boost circuit to control operations of the battery charging switch and the battery discharging switch such that the bidirectional buck-boost circuit is configured to selectively conduct charging current from the rectifier to the battery and selectively conduct welding current from the battery to the switching type power converter.

[0006] In accordance with another aspect of the present invention, provided is a hybrid arc welding system. The hybrid arc welding system includes a welding waveform generator comprising a switching type power converter. A welding torch is operatively connected to the switching type power converter to receive electrical energy from the switching type power converter and produce an electric arc for welding. The system further includes an engine-generator and a rectifier connected to receive an output of the engine-generator and supply electrical energy to the switching type power converter over a DC bus. A filtering capacitor is connected across the DC bus at an output of the rectifier. A battery is connected to receive an output of the rectifier and supply electrical energy to the switching type power converter. A bidirectional buck-boost circuit comprises a first electronic switch, and an inductor electrically connected between the battery and the first electronic switch. The first electronic switch is electrically connected between the inductor and a positive terminal of the DC bus. The bidirectional buck-boost circuit also includes a second electronic switch electrically connected between a negative terminal of the DC bus and both of the inductor and the first electronic switch. A controller is operatively connected to the bidirectional buck-boost circuit to control operations of the first electronic switch and the second electronic switch such that the bidirectional buck-boost circuit is configured to selectively conduct charging current from the rectifier to the battery and selectively conduct welding current from the battery to the switching type power converter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The foregoing and other aspects of the invention will become apparent to those skilled in the art to which the invention relates upon reading the following description with reference to the accompanying drawings, in which:

[0008] FIG. 1 is a perspective view of an engine driven welder hybrid arc welding system;

[0009] FIG. 2 is a schematic diagram of an engine driven welder hybrid arc welding system;

[0010] FIG. 3 is a schematic diagram of an engine driven welder hybrid arc welding system;

[0011] FIG. 4 is a schematic diagram of an engine driven welder hybrid arc welding system;

[0012] FIG. 5 is a schematic diagram of an engine driven welder hybrid arc welding system;

[0013] FIG. 6 is a schematic diagram of an engine driven welder hybrid arc welding system; and

[0014] FIG. 7 illustrates an example controller.

DETAILED DESCRIPTION OF THE INVENTION

[0015] The present invention relates to a hybrid arc welding system having both an engine driven generator and a battery for supplying welding current to a welding torch. The present invention will now be described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. It is to be appreciated that the various drawings are not necessarily drawn to scale from one figure to another nor inside a given figure, and in particular that the size of the components are arbitrarily drawn for facilitating the understanding of the drawings. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It may be evident, however, that the present invention can be practiced without these specific details. Additionally, other embodiments

of the invention are possible and the invention is capable of being practiced and carried out in ways other than as described. The terminology and phraseology used in describing the invention is employed for the purpose of promoting an understanding of the invention and should not be taken as limiting.

[0016] As used herein, “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together. Any disjunctive word or phrase presenting two or more alternative terms, whether in the description of embodiments, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” should be understood to include the possibilities of “A” or “B” or “A and B.”

[0017] While embodiments of the present invention described herein are discussed in the context of a gas metal arc welding (GMAW) system, other embodiments of the invention are not limited thereto. For example, embodiments can be utilized in shielded metal arc welding (SAW), flux-cored arc welding (FCAW), metal-cored arc welding (MCAW), gas tungsten arc welding (GTAW) as well as other similar types of welding operations. Further, embodiments of the present invention can be used in manual, semi-automatic and robotic welding operations. Embodiments of the present invention can also be used in metal deposition operations that are similar to welding, such as additive manufacturing, hardfacing, and cladding. As used herein, the term “welding” is intended to encompass all of these technologies as they all involve material deposition to either join or build up a workpiece. Therefore, in the interests of efficiency, the term “welding” is used below in the description of exemplary embodiments, but is intended to include all of these material deposition operations, whether or not joining of multiple workpieces occurs.

[0018] FIG. 1 is an outer perspective view of an example engine driven hybrid arc welding machine **10** or system. The welding machine **10** can include a base **12** that is part of a chassis structure for the welding machine, and an outer case or enclosure **14**. A user interface **16** for controlling the operation of the hybrid arc welding machine **10** is located on the front side of the outer case **14**. Example welding processes that can be performed by the welding machine **10** include shielded metal arc welding (SMAW), gas metal arc welding (GMAW), flux-cored arc welding (FCAW), gas tungsten arc welding (GTAW), and gouging. Within the outer case **14** is an internal combustion engine that drives a generator, a hybrid battery, and a welding waveform generator (e.g., welding power circuitry) having a switching type power converter, such as an inverter or a chopper.

[0019] FIG. 2 is a schematic diagram of a hybrid arc welding system, which is an engine driven welding system having a battery to provide additional welding energy as needed. The welding system includes a generator **30** driven by an engine **28** thereby forming an engine-generator. Example engines include diesel engines, gasoline engines, LP gas engines, and the like. The generator **30** generates electrical energy for powering a welding power supply or welding waveform generator having a switching type power converter **34** (e.g., a chopper or inverter). The generator **30** can be a synchronous 3-phase alternator. However, the generator need not be a synchronous 3-phase alternator. For example, the generator could be a single phase alternator or a DC generator if desired.

[0020] The output of the generator **30** is rectified to DC by a rectifier **32**. The rectifier **32** supplies DC power to a DC bus **46** that conducts electricity to the switching type power converter **34**. Thus, the engine-driven generator **30** is operatively connected to the welding waveform generator having the switching type power converter **34** (e.g., a chopper or inverter), and supplies electrical energy to the welding waveform generator to generate a welding arc **36**. The switching type power converter **34** converts the electrical energy from the generator **30** to a welding output for generating

an electric welding arc **36** between a torch **38** and workpiece(s) **40** to be welded. Thus, the welding torch **38** is operatively connected to the switching type power converter **34** to receive electrical energy from the switching type power converter and produce the electric arc **36** for welding. The switching type power converter **34** can include a chopper or inverter circuitry for generating the welding output, control circuitry for controlling the chopper or inverter circuitry, and may include a transformer and one or more rectifiers. The specific construction of welding waveform generators or welding power supplies having switching type power converters such as choppers or inverters is well known to one of ordinary skill in the art and will not be discussed in detail herein.

[0021] The hybrid arc welding system also includes a battery **42** (e.g., a lithium ion battery) that is connected to the DC bus **46** through a charge-discharge circuit **44**. The battery **42** can be a single battery or a bank of batteries as is known in the art. The battery **42** is connected to the DC bus **46** and to the charge-discharge circuit **44** to receive an output of the rectifier **32** for charging the battery and to supply electrical energy to the switching type power converter **34**. The battery **42** receives charging/recharging power from the generator **30** and rectifier **32** through the charge-discharge circuit **44**. The battery **42** also supplies welding current to the switching type power converter **34** through the charge-discharge circuit **44**. The battery **42** can supplement the engine **28** and generator **30** when supplying power for welding and other auxiliary operations, or the battery can supply power for welding and/or auxiliary operations by itself. The battery **42** is connected such that it provides power to the torch **38** and/or an auxiliary output independently of the generator **30** or with the generator. In this sense, power can be provided by at least one of the engine-driven generator **30** and the battery **42** such that power sufficient to perform an operation may be provided by generator or battery independently, or the battery may supplement power from the generator.

[0022] In conventional hybrid arc welding systems, the charge/discharge circuit **44** is not bidirectional. For example, the charge/discharge circuit **44** can include separate buck and boost subcircuits to charge the battery **42** and discharge the battery, respectively. Alternatively and desirably, the charge-discharge circuit **44** can be a bidirectional battery charge-discharge circuit (e.g., a bidirectional buck-boost circuit) that combines the buck and boost subcircuits to minimize the number of components in the charge/discharge circuit, thereby reducing the cost of the system and the space footprint of the charge/discharge circuit.

[0023] FIG. **3** is a schematic diagram of a hybrid arc welding system without a bidirectional charge-discharge circuit. Hybrid battery **42** is charged from the DC bus **46** through a buck subcircuit **50**. The buck subcircuit **50** can include an electronic switch, such as a transistor switch, and a diode and inductor as shown. The electronic switch in the buck subcircuit **50** operates to charge the battery **42** at a lower voltage than the DC bus **46** when the engine **28** is running and there is sufficient spare capacity from the generator **30** (e.g., no welding is occurring). The battery **42** is discharged to supply welding power to the DC bus **46** through a boost subcircuit **52**. The boost subcircuit **52** can include a further electronic switch and an inductor and diode as shown. The electronic switch in the boost subcircuit **52** operates to discharge the battery **42** and raise its voltage to the DC bus **46** voltage. The welding system can include one or more controllers **54**. The controller **54** controls operations of the switching type power converter (shown in FIG. **3** as a welding chopper) to generate a desired welding waveform, using pulse width modulation control for example. The controller **54** can also control operations of the buck and boost subcircuit **50**, **52** electronic switches, or there can be a separate dedicated controller for the subcircuits. In either case, the operations of the charge-discharge circuit can be controlled based on the charge level of the battery **42**, the load on the engine-generator, the desired amount of welding current, etc., which can be monitored and/or determined by the controller **54**.

[0024] FIG. **4** is a schematic diagram of a hybrid arc welding system having a bidirectional charge-discharge circuit or bidirectional buck-boost circuit **56**. In FIG. **3**, charging and discharging of the battery **42** is achieved via separate subcircuits with separate inductors/chokes, electronic switches,

and current paths. However, the bidirectional buck-boost circuit **56** in FIG. **4** has a combined current path utilizing battery charging and discharging electronic switches **58**, **60** and a single inductor **62**. The battery charging switch **58** is directly connected to the DC bus **46** (e.g., the positive rail of the DC bus), the filter capacitor **64**, and the battery discharging switch **60**. The battery discharging switch **60** is directly connected to the battery charging switch **58**, inductor **62** (battery choke), and the DC bus **46** (e.g., the negative rail of the DC bus). The inductor **62** is electrically connected between the battery **42** and the connection of the charging switch **58** to the discharging switch **60**. Compared to the circuitry in FIG. **3**, the bidirectional charge-discharge circuit or bidirectional buck boost circuit **56** in FIG. **4** has fewer components that take up less space. Moreover, efficiency can be improved by employing synchronous rectification using the electronic switches **58**, **60**.

[0025] In the example embodiment shown in FIG. **4**, the battery charging and discharging switches **58**, **60** can be electronic switches, such as transistor switches. For example, the electronic switches could be power transistors, IGBTs, MOSFETs, etc. Each electronic switch could have a separate diode electrically connected in parallel with the electronic switch as shown in FIG. **6**. However, if MOSFETs are used as shown in FIGS. **4** and **5**, synchronous rectification can be performed utilizing the intrinsic body diodes to improve efficiency. Thus, if desired, the battery charging and discharging switches can lack the respective separate diodes connected in parallel that are shown in FIG. **6**. The intrinsic body diodes of the battery charging and discharging switches **58**, **60** are shown in FIG. **4** and are not shown in FIG. **5**.

[0026] With reference to FIG. **4**, the bidirectional buck-boost circuit **56** is connected to the DC bus **46** between the rectifier **32** and the switching type power converter **34**. To charge the battery **42** from the generator **30**, the gate of the battery charging switch **58** is controlled (e.g., via pulse width modulation or PWM, pulse frequency modulation, and the like) to create a charging current flow to the battery. If synchronous rectification is not being performed and the battery discharging switch **60** has a separate parallel diode, then the battery discharging switch can remain off during battery charging. However, if synchronous rectification is being performed, then the battery discharging switch **60** will be activated via its gate when the battery charging switch **58** is turned off during PWM and after a deadband (to prevent both switches being activated simultaneously).

[0027] To discharge the battery **42** to supply power to the DC bus **46**, the gate of the battery discharging switch **60** is controlled (e.g., via PWM) to create a discharging current flow out of the battery. If synchronous rectification is not being performed and the battery charging switch **58** has a separate parallel diode, then the battery charging switch can remain off during battery discharging. However, if synchronous rectification is being performed, then the battery charging switch **58** will be activated via its gate when the battery discharging switch **60** is turned off during PWM and after a deadband (to prevent both switches being activated simultaneously).

[0028] It is to be appreciated that the duty cycles of the battery charging and discharging switches **58**, **60** control the energy to the inductor **62** or battery choke. When charging the battery **42**, a shorter PWM duty cycle for the battery charging switch **58** will provide a greater decrease in voltage from the DC bus **46**. When discharging the battery **42**, a longer PWM duty cycle for the battery discharging switch **60** will provide a greater boost in voltage from the battery to the DC bus **46**.

[0029] The bidirectional buck-boost circuit **56** can include a filtering capacitor **64** connected across the output of the rectifier **32** that prevents surges from over voltage conditions on the battery charging and discharging switches **58**, **60** and provides extra filtering from AC.

[0030] As noted above, the welding system can include one or more controllers. The controller **54** controls operations of the switching type power converter **34** (shown in FIG. **4** as a welding chopper but it could be an inverter) to generate a desired welding waveform, using PWM control for example. The controller **54** can also control operations of the bidirectional charge-discharge circuit or bidirectional buck-boost circuit **56**, in particular the operations of the charge and

discharge electronic switches **58, 60** in the charge-discharge circuit. However, in the example embodiment of FIG. **4**, the welding system includes a separate controller **66** for controlling the operations of the charge and discharge electronic switches **58, 60** of the bidirectional charge-discharge circuit or bidirectional buck-boost circuit **56**. The controller **66** is operatively connected to the bidirectional battery charge-discharge circuit or bidirectional buck-boost circuit to control operations of the charge and discharge electronic switches **58, 60** such that the bidirectional battery charge-discharge circuit or bidirectional buck-boost circuit is configured to selectively conduct charging current from the rectifier **32** to the battery **42** and selectively conduct welding current from the battery to the switching type power converter **34**. The controller **66** for the bidirectional buck-boost circuit **56** can provide PWM and synchronous rectification signals to the charge and discharge electronic switches **58, 60** to provide battery **42** charging and discharging current at appropriate voltage levels. The operations of the bidirectional charge-discharge circuit or bidirectional buck-boost circuit **56** can be controlled based on the charge level of the battery **42**, the load on the engine-generator, the desired amount of welding current, etc., which can be monitored and/or determined by one or both of the controllers **54, 66**. In certain embodiments, the controller **54** for the switching type power converter **34** can monitor parameters such as engine RPM, welding voltage level, welding current level, DC bus voltage level, etc. and communicate power or PWM adjustments to the controller **66** for the bidirectional buck-boost circuit **56** to increase or reduce charging current and/or discharge current from the hybrid battery **42**. For example, the controllers **54, 66** can communicate bidirectionally via a CAN bus for example. The controller **54** for the switching type power converter **34** can instruct the controller **66** for the bidirectional buck-boost circuit **56** to increase and/or decrease power to or from the hybrid battery **42**, and the controller for the bidirectional buck-boost circuit **56** can control the operation and synchronous rectification of the charge and discharge electronic switches **58, 60** accordingly. PWM adjustments for the charge and discharge electronic switches **58, 60** can occur dynamically during operation of the generator **30** and/or during a welding operation.

[0031] FIG. **5** is a schematic diagram of a hybrid arc welding system having a bidirectional charge-discharge circuit or bidirectional buck-boost circuit **56** in which the charge and discharge electronic switches **58a, 60a** are MOSFETs that lack separate diodes connected in parallel. The controller **66** for the bidirectional buck-boost circuit **56** controls the operation of the switches **58a, 60a** to perform synchronous rectification utilizing the intrinsic body diodes (not shown) of the MOSFET switches. The switches **58a, 60a** are operated together during battery charging and discharging with deadbands between activation of the switches to prevent both switches being activated simultaneously. FIG. **6** is a schematic diagram of a hybrid arc welding system having a bidirectional charge-discharge circuit or bidirectional buck-boost circuit **56** in which the charge and discharge electronic switches **58a, 60a** are MOSFETs that include separate diodes **59, 61** (separate from the intrinsic body diodes) connected in parallel with the switches.

[0032] The hybrid arc welding systems discussed herein can include auxiliary outputs for powering equipment such as lights, grinders, various plug-in hand tools, etc. In certain embodiments, the hybrid arc welding system can include an inverter connected to the hybrid battery through a contactor to provide auxiliary power output.

[0033] FIG. **7** illustrates an example embodiment of controller **800** for use in the example hybrid arc welding system discussed above (e.g., a controller for a switching type power converter and/or a control for hybrid battery charge-discharge circuitry.) The example controller **800** includes at least one processor **814** which communicates with a number of peripheral devices via bus subsystem **812**. These peripheral devices may include a storage subsystem **824**, including, for example, a memory subsystem **828** and a file storage subsystem **826**, user interface input devices **822**, user interface output devices **820**, and a network interface subsystem **816**. The input and output devices allow user interaction with the controller **800**. Network interface subsystem **816** provides an interface to outside networks and can be coupled to corresponding interface devices in

other computer systems or programmable devices.

[0034] User interface input devices **822** may include a keyboard, pointing devices such as a mouse, trackball, touchpad, or graphics tablet, a scanner, a touchscreen incorporated into the display, audio input devices such as voice recognition systems, microphones, and/or other types of input devices. In general, use of the term “input device” is intended to include all possible types of devices and ways to input information into the controller **800** or onto a communication network.

[0035] User interface output devices **820** may include a display subsystem, a printer, a fax machine, or non-visual displays such as audio output devices. The display subsystem may include a cathode ray tube (CRT), a flat-panel device such as a liquid crystal display (LCD), a projection device, or some other mechanism for creating a visible image. The display subsystem may also provide non-visual display such as via audio output devices. In general, use of the term “output device” is intended to include all possible types of devices and ways to output information from the controller **800** to the user or to another machine or computer system.

[0036] Storage subsystem **824** stores programming and data constructs that provide the functionality of some or all of the operations described herein. For example, the storage subsystem **824** may include programming instructions to allow the controller **800** to execute the PID routine and the consumable identification described above.

[0037] Firmware or software modules having the programming instructions are generally executed by processor **814** alone or in combination with other processors. Memory subsystem **828** used in the storage subsystem **824** can include a number of memories including a main random access memory (RAM) **830** for storage of instructions and data during program execution and a read only memory (ROM) **832** in which fixed instructions are stored. A file storage subsystem **826** can provide persistent storage for program and data files, and may include a hard disk drive, a floppy disk drive along with associated removable media, a CD-ROM drive, an optical drive, or removable media cartridges. The modules implementing the functionality of certain embodiments may be stored by file storage subsystem **826** in the storage subsystem **824**, or in other machines accessible by the processor(s) **814**.

[0038] Bus subsystem **812** provides a mechanism for letting the various components and subsystems of the controller **800** communicate with each other as intended. Although bus subsystem **812** is shown schematically as a single bus, alternative embodiments of the bus subsystem may use multiple buses.

[0039] It should be evident that this disclosure is by way of example and that various changes may be made by adding, modifying or eliminating details without departing from the fair scope of the teaching contained in this disclosure. The invention is therefore not limited to particular details of this disclosure except to the extent that the following claims are necessarily so limited.

Claims

1. A hybrid arc welding system, comprising: a welding waveform generator comprising a switching type power converter; a welding torch operatively connected to the switching type power converter to receive electrical energy from the switching type power converter and produce an electric arc for welding; an engine-generator; a rectifier connected to receive an output of the engine-generator and supply electrical energy to the switching type power converter; a battery connected to receive an output of the rectifier and supply electrical energy to the switching type power converter; a bidirectional battery charge-discharge circuit connected between the rectifier and the switching type power converter, wherein the bidirectional battery charge-discharge circuit comprises a first electronic switch and a second electronic switch; and a controller operatively connected to the bidirectional battery charge-discharge circuit to control operations of the first electronic switch and the second electronic switch such that the bidirectional battery charge-discharge circuit is configured to selectively conduct charging current from the rectifier to the battery and selectively

conduct welding current from the battery to the switching type power converter.

2. The hybrid arc welding system of claim 1, wherein the first electronic switch and the second electronic switch are MOSFET switches.

3. The hybrid arc welding system of claim 2, wherein the controller controls operations of the first electronic switch and the second electronic switch to perform synchronous rectification.

4. The hybrid arc welding system of claim 1, further comprising a first diode electrically connected in parallel with the first electronic switch, and a second diode electrically connected in parallel with the second electronic switch.

5. The hybrid arc welding system of claim 1, further comprising a filtering capacitor connected across the output of the rectifier.

6. The hybrid arc welding system of claim 5, wherein the bidirectional battery charge-discharge circuit further comprises an inductor electrically connected between the battery and a connection of the first electronic switch to the second electronic switch.

7. The hybrid arc welding system of claim 1, wherein the first electronic switch is directly connected to the second electronic switch.

8. A hybrid arc welding system, comprising: a welding waveform generator comprising a switching type power converter; a welding torch operatively connected to the switching type power converter to receive electrical energy from the switching type power converter and produce an electric arc for welding; an engine-generator; a rectifier connected to receive an output of the engine-generator and supply electrical energy to the switching type power converter; a battery connected to receive an output of the rectifier and supply electrical energy to the switching type power converter; a bidirectional buck-boost circuit connected between the rectifier and the switching type power converter, wherein the bidirectional buck-boost circuit comprises a battery charging switch without a separate diode connected in parallel with the battery charging switch, and a battery discharging switch without a separate diode connected in parallel with the battery discharging switch; and a controller operatively connected to the bidirectional buck-boost circuit to control operations of the battery charging switch and the battery discharging switch such that the bidirectional buck-boost circuit is configured to selectively conduct charging current from the rectifier to the battery and selectively conduct welding current from the battery to the switching type power converter.

9. The hybrid arc welding system of claim 8, wherein the battery charging switch and the battery discharging switch are MOSFET switches.

10. The hybrid arc welding system of claim 8, wherein the controller controls operations of the battery charging switch and the battery discharging switch to perform synchronous rectification.

11. The hybrid arc welding system of claim 8, further comprising a filtering capacitor connected across the output of the rectifier.

12. The hybrid arc welding system of claim 11, wherein the bidirectional buck-boost circuit further comprises an inductor electrically connected between the battery and a connection of the battery charging switch to the battery discharging switch.

13. The hybrid arc welding system of claim 8, wherein the battery charging switch is directly connected to the battery discharging switch.

14. A hybrid arc welding system, comprising: a welding waveform generator comprising a switching type power converter; a welding torch operatively connected to the switching type power converter to receive electrical energy from the switching type power converter and produce an electric arc for welding; an engine-generator; a rectifier connected to receive an output of the engine-generator and supply electrical energy to the switching type power converter over a DC bus; a filtering capacitor connected across the DC bus at an output of the rectifier; a battery connected to receive an output of the rectifier and supply electrical energy to the switching type power converter; a bidirectional buck-boost circuit comprising: a first electronic switch; an inductor electrically connected between the battery and the first electronic switch, wherein the first electronic switch is electrically connected between the inductor and a positive terminal of the DC

bus, and a second electronic switch electrically connected between a negative terminal of the DC bus and both of the inductor and the first electronic switch, and a controller operatively connected to the bidirectional buck-boost circuit to control operations of the first electronic switch and the second electronic switch such that the bidirectional buck-boost circuit is configured to selectively conduct charging current from the rectifier to the battery and selectively conduct welding current from the battery to the switching type power converter.

15. The hybrid arc welding system of claim 14, wherein the first electronic switch and the second electronic switch are MOSFET switches.

16. The hybrid arc welding system of claim 15, wherein the controller controls operations of the first electronic switch and the second electronic switch to perform synchronous rectification.

17. The hybrid arc welding system of claim 14, further comprising a first diode electrically connected in parallel with the first electronic switch, and a second diode electrically connected in parallel with the second electronic switch.
